# US COVID-19 Map and Network Visualization

*Testing whether geographic proximity predicts COVID-19 case counts via a state-similarity network, community detection, and an interactive heatmap.*

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## Overview

This project investigates whether US states that are geographically close also share similar COVID-19 case and death counts. Using daily county-level COVID-19 data, it aggregates totals per state, constructs a state-similarity network via a Gaussian affinity kernel, runs Louvain community detection, and generates a geographic heatmap. The analysis covers all 50 states and uses NetworkX for graph construction and folium for interactive mapping.


*State-similarity network derived from COVID-19 trends.*

## Goals

Determine whether spatial proximity between US states is a meaningful predictor of COVID-19 case magnitude, and visualize the resulting community structure both as a network graph and as a geographic heatmap.

## Objectives

- Compute total confirmed cases (TNC) and total deaths (TND) per state by summing county-level daily data.
- Build a Gaussian affinity matrix from each state's (TNC, TND) vector and threshold it into an adjacency matrix at a value of 0.5.
- Construct and visualize the state-similarity graph using NetworkX, with nodes labelled by state initials.
- Apply Louvain community detection and repeat the analysis across multiple values of the kernel parameter sigma to assess stability.

- Export the graph to GEXF format for Gephi and produce an interactive geographic heatmap using folium.

## Approach

State-level totals are derived from the latest cumulative county counts. A Gaussian kernel over the (TNC, TND) vector pairs encodes numerical similarity as edge weights; thresholding at 0.5 yields a binary adjacency matrix. The graph is built with NetworkX and communities are detected via the Louvain algorithm. Sigma is varied to probe how sensitive the community structure is to the kernel bandwidth. The final heatmap is rendered with folium to overlay case density on actual state geography.

## Outcomes

- Community structure grouped states by case and death magnitude rather than by geographic location, contradicting the initial hypothesis.
- The highest-population states — California, Texas, Florida, and New York — remained isolated nodes because no other state was numerically close to them.
- Neighbouring states frequently belonged to different communities while geographically distant states with similar population sizes shared the same community.
- The evidence indicates that population size is a far stronger predictor of COVID-19 case counts than spatial proximity.
- The interactive heatmap and GEXF export provide shareable artefacts for further exploration in folium and Gephi respectively.